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SUBMITTING STUDENT		
SURNAME ROY	GIVEN NAMES RISHIRAJ	STUDENT NUMBER 23921699
UNIT NAME ENGINEERING RESEARCH PROJECT 1		UNIT CODE GENG4411
TITLE/TOPIC OF ASSIGNMENT PROJECT PROPOSAL		NAME OF LECTURER/TUTOR KIERAN SCOTT QUIRKE-BROWN
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GENG4411 Engineering Research Project Part 1

Project Proposal

Low Altitude Onboard Detection Pipeline for Person and Human Presence Indicators Under Vegetative Occlusion

Author: Rishiraj Roy

Student Number: 23921699

School of Engineering, University of Western Australia

Supervisor: Dr Kieran Scott Quirke-Brown

School of Engineering, University of Western Australia

Project Type: Design and Build

Word count: 3975

School of Engineering
University of Western Australia

Date: 20th September 2026

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Abstract

Search and Rescue (SAR) efforts over forested terrain for a lost person is highly time critical and aerial detectors that perform well over open terrain usually fail once occlusion and bush canopy cover hides the target. This project builds a low altitude (5 m - 10 m) detection pipeline running onboard a Jetson Orin Nano Super and finds both missing persons and human presence indicator (HPI) classes like backpacks, shelters and discarded clothing. It is hypothesised that recall and mAP@50 degrade gradually rather than fully collapsing as occlusion increases.

This thesis project compares six detector architectures, spanning over convolution, attention, hypergraph correlation and transformer mechanisms, as an elimination tournament on accuracy and onboard latency and memory. Performance measurements are under a frozen, graded occlusion sweep from 0 to 80% mask levels and cross checked on real bush imagery. Vision language model (VLMs) and open vocabulary detectors (OVDs) are tested after as escalation stages for low confidence detections and for HPI classes that are purposely withheld from training, and are kept only if they beat cheap re-detection of cropped regions. Coordinated SAR by a drone fleet is evaluated in simulation with real world testing to follow, dependent on full system builds being completed by teammates.

This thesis tests how far RGB detection by itself can be pushed under occlusion within an edge compute budget while also treating a negative result for the language model stages as valid finding for future work.

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Nomenclature

Abbreviations

ARM64	64-bit ARM processor architecture; the Jetson target, as opposed to the x86 training workstation.
CNN	Convolutional Neural Network.
COCO	Common Objects in Context; general-purpose dataset used for detector pretraining.
CUDA	Compute Unified Device Architecture; NVIDIA's GPU computing platform.
CVAT	Computer Vision Annotation Tool; used to annotate the bush dataset.
DFL	Distribution Focal Loss; box regression component removed in YOLO26.
FOV	Field of View.
fps	Frames per second.
GPU	Graphics Processing Unit.
HPI	Human Presence Indicator; an object or trace showing a person has been present (e.g. backpack, discarded clothing, shelter).
INT8	8-bit integer precision used for quantised inference.
LoRA	Low-Rank Adaptation; parameter-efficient fine-tuning method designed for transformer layers.
NMS	Non-Maximum Suppression; removes duplicate detections.
NVMe SSD	Non-Volatile Memory express Solid State Drive.
OVD	Open Vocabulary Detector; prompted with class names as text.
PX4	Open-source flight control software (autopilot).
R-CNN	Region-based Convolutional Neural Network; two-stage detector family (Faster R-CNN).
RGB	Red, Green, Blue; standard visible-light imagery.
ROS 2	Robot Operating System 2.
SAR	Search and Rescue.
SITL	Software-In-The-Loop; autopilot run in simulation without flight hardware.
TensorRT	NVIDIA inference optimisation library and runtime.
TOPS	Tera Operations Per Second; manufacturer-claimed peak compute, not measured throughput.
UAV	Unmanned Aerial Vehicle.
uXRCE-DDS	Micro eXtremely Resource Constrained Environment - Data Distribution Service; middleware that bridges PX4 and ROS 2.
VLM	Vision Language Model.
YOLO	You Only Look Once; single-stage real-time object detector family.

Metrics and Symbols

TP, FP, FN	True positives, false positives and false negatives.
P	Precision, $P = TP / (TP + FP)$.
R	Recall, $R = TP / (TP + FN)$.
IoU	Intersection over Union between a predicted and a ground truth box.
AP, mAP	Average Precision for one class (area under the precision versus recall curve); mean Average Precision is AP averaged over classes.
mAP@50	mAP with a detection counted correct at $IoU \geq 0.50$.
mAP@50–95	mAP averaged over IoU thresholds 0.50 to 0.95 in steps of 0.05.

Project Terms

Baseline A, B	Baseline detectors trained on HERIDAL (A, person) and Weitefeld (B, shelter, object and person).
Finding	In Weitefeld, one physical confirmed discovery; the unit used to split train, validation and test so an object never appears in two splits.
Held-out class	HPI class withheld from all training, used only to test VLM and OVD generalisation.
HERIDAL	Aerial SAR person detection dataset captured at 45 m - 60 m over open terrain.
Pareto front	Set of model configurations where accuracy cannot improve without worse latency or memory use.
Pre-fill	VLM stage that encodes image and prompt tokens before generating output; dominates onboard VLM latency.
Re-detection	Cropped, higher-resolution re-run of the detector on low-confidence regions; the cheap baseline a VLM stage must beat.
Tile	Fixed size image crop cut from a large source photograph (HERIDAL: 1024 px, 0.2 overlap).
Weitefeld	Real SAR operation dataset over dense forest, captured from a crewed aircraft at ~ 300 m.

1 Introduction

Search and rescue for a lost person in vegetative terrain is time critical. As time passes the search area grows and survival rate decreases [6]. Over open terrain, aerial imagery already assists ground search, but bushland canopy occlusion hides both people and traces of people that responders rely on, which matters to police, SAR agencies, volunteer searchers and families of the missing people [7].

HERIDAL based systems perform well at 45 m - 60 m over open terrain with a 88.9% detection rate [1]. This result is not transferrable to canopy; on real forest search scenery captured at ~ 300 m, a modern YOLO detector model effectively failed [3], with detectors trained on standing, unoccluded subjects degrade sharply once a subject's visibility or pose differs from the training distribution [8]. The root issue across all three results only a few reliable pixels on partially hidden targets, compounded by an onboard compute budget that limits the real time recovery a detector can achieve on fragmented evidence.

This project is aimed at the gap directly, by proposing an onboard, 5 m - 10 m low altitude RGB pipeline running on a Jetson Orin Nano Super, evaluated under real and graded synthetic occlusion, instead of a single accuracy value, with human presence indicators parallel to person detection as the primary contribution. It also proposes a fair comparison against open vocabulary detector and visual language model based re-detection stages under the same compute budget. Multi drone coordination is evaluated as a secondary objective where this UAV works in tandem with thesis team's airframes in a drone fleet for optimised SAR.

2 Literature Review

Aerial imagery is stock standard in SAR with detection over open terrain being mostly solved. HERIDAL based systems find people in unobstructed environments seen from 45 m - 60 m [1] with vegetated terrain being quite the opposite. A modern generation detector failed on real search datasets captured at ~ 300 m [3], with recent surveys naming occlusion an unsolved problem in aerial person detection [7]. Studies show raw inference onboard embedded software ran at 2 fps [9], thus robustness must be introduced within a fixed compute confinement for onboard inferences. This review uncovers that lineage, from open terrain SAR, into occluded canopy and identifies what remains untested.

2.1 Aerial Based Person Detection in Open Terrain

Lineage of detectors used in Aerial SAR have followed general trends in detector architecture evolution, starting with two stage detectors like Faster R-CNN [10], adapted by HERIDAL for SAR imagery [1]. Currently, single stage YOLO variants are the dominant choice for real time operation [11], [9]. Recent studies categorise current methods into four groups (sample oriented sparse, information fusion, on device lightweight and scale and perspective aware) and puts occlusion as a challenge unsolved across all four [7].

HERIDAL achieves a recall of 88.9% with 34.8% precision on unoccluded open terrain scenes at 45 m - 60 m [1] with Ciccone and Ceruti's best YOLOv5s-PBfpn-Deconv model reaching 0.802 mAP@50 using the same dataset [9], where neither figure is successfully transferred to canopy cover.



Figure 1: Zoomed and native resolution HERIDAL scene of two people. Even with 88.9% recall in the dataset, magnification is required on targets to located against mixed open terrain and tree cover. Adapted from [1]

Reported speed is also hardware dependent where Faster R-CNN runs at 5 - 17 fps on a workstation[10], whereas a YOLOv5s pipeline reached ~ 2 fps on a Jetson Nano without INT8 or TensorRT optimisation [9]. On a custom video dataset, YOLO11n/s achieve to have mAP@50 between 0.73 - 0.78 and mAP@50-95 of 0.29 - 0.31 as reported by Botea et al. Imprecise bounding boxes are penalised more harshly by mAP@50-95 than mAP@50 does. This gap points away from an outright target detection weakness towards a localisation one [11].

2.2 Vegetative Occlusion

When SAR detectors are trained on unobstructed, standing people, they sharply degrade on partially occluded, non standing objects. Recent occlusion based datasets built specifically for this gap have conventional models missing targets differing from their pose and visibility while training [8]. One sensor based solution is RGB thermal fusion, covering detections through sparse foliage by using thermal contrast that a RGB based model can't see [12]. The result is promising but assumes the use of a spatially aligned sensor alongside the RGB sensor and the calibration required to run simultaneously. This project asks how far a single RGB channel goes before the extra sensor is a necessity, hence RGB thermal fusion is out of scope here.

Synthetic occlusion is the established alternative when paired sensor data doesn't exist. Occluding a controlled fraction of an image while training is a standard evaluation and regularization tool for convolutional detectors [13]. There are risks associated with this. A detector could learn that occluded texture implies a target rather than a target is present under occlusion, if the masks only appear above real subjects, which motivates pasting identical occluders over background regions during training as a controlled variable. Synthetic occlusions are also texturally and geometrically more rudimentary, so a degradation curve for synthetic masks is a lower bound on difficulty in the real world rather than substituting it.

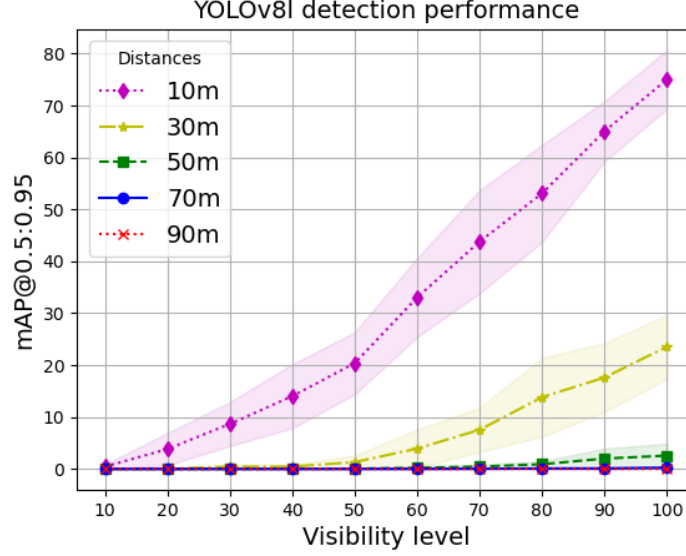


Figure 2: YOLOv8l detection performance vs NOMAD visibility level, plotted by distance. Adapted from [2]

This difficulty is directly mapped by Figure 2. On NOMAD’s graded visibility aerial scenery, the $mAP@0.5:0.95$ fails toward zero further than ~ 50 m at all visibility levels and has a steep collapse even at 10 m when visibility is lower than 60 [2]. Consequently, this motivates a graded synthetic occlusion sweep, cross-referenced against a real occluded dataset, allowing results to hold outside of simulation. A full box label policy also scores presence detection rather than visible area.

2.3 Altitude Determined Detection Failure

Weitefeld is a real SAR operation dataset with 10659 images, 34424 bounding boxes and 405 findings confirmed across four classes, taken over dense forest cover [3]. A modern detector, YOLOv12 failed on this dataset, reported at confidence of $\sim 0.016\%$, attributed as too few subject pixels under partially occluded clues at an altitude of 300 m and a sample distance at ground of 3 cm - 5 cm per pixel.

There are several limitations when it comes to generalising. The imagery doesn’t come from an UAV, rather a crewed aircraft. The four dataset classes are also coarse with labels being subjective and crowd sourced with a messy ‘unknown’ class. This project’s own examination found label noise in the person class.

Nothing here addresses the low altitude flight of this project, a finer HPI taxonomy, VLM/OVD detection methods or multi drone coordination. The relevance of Weitefeld is as independent evidence for low altitude premise since it shows a detector failing at 300 m on a real canopy and like a real multi class dataset which this project’s own multi class baseline will be checked.

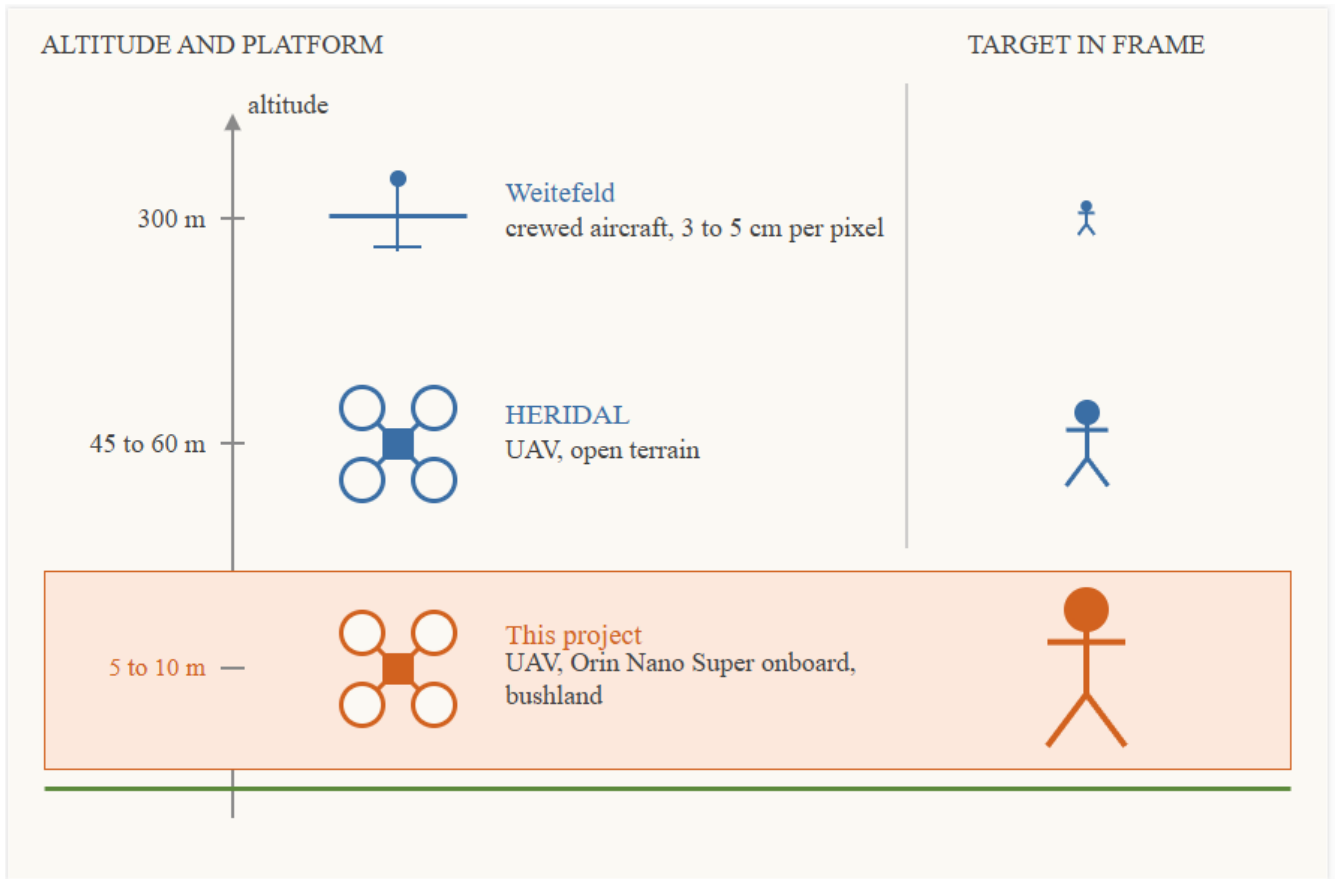


Figure 3: Operational altitude of Weitefeld [3], HERIDAL[1] and this project’s proposed system. Not to scale, stick figures show relative size of target in frame. Constructed on Draw.io software

2.4 Detector Architectures for Occluded Small Targets

This project’s detector sweep is anchored by six architectures, chosen for structural differences in their underlying mechanisms rather than having six variants of the same idea. YOLO11 is a plain convolutional detector with only local convolution kernel [14], and serves as a continuity reference at zero mechanism cost. YOLO26 maintains the convolutional backbone, however strips the inference time overhead with no DFL or NMS step, making it suitable for edge deployment [14]. It still carries the same local convolutions only limitation but with lower latency than YOLO11.

YOLOv12 adds attention within the CNN but limited to local regions and not the whole image [15]. Local attention can re-weight neighbouring evidence, which can help when visible fragments of a target sit close together. Since that attention is confined to local, so it may not link evidence across a wide occluder which the occlusion sweep tests.

YOLOv13 uses hypergraph correlation that replaces the local mechanism by letting scattered, non adjacent regions of an image connect directly [16]. This is a structural fit for partial occlusion where only visible clues may be scattered pixels with no local neighbourhoods. YOLOv13 is also the only one of the six architectures with its paper not yet peer reviewed, so its results should be reported with that limitation in mind, not treated equally as the other five.

RF-DETR and D-FINE are full transformer based detectors. They replace convolution with global self attention across the whole image [17], [18]. Global attention is the direct solution to fragmented evidence, but transformers need a lot of data and have more export risk to the embedded hardware of the target than a CNN.

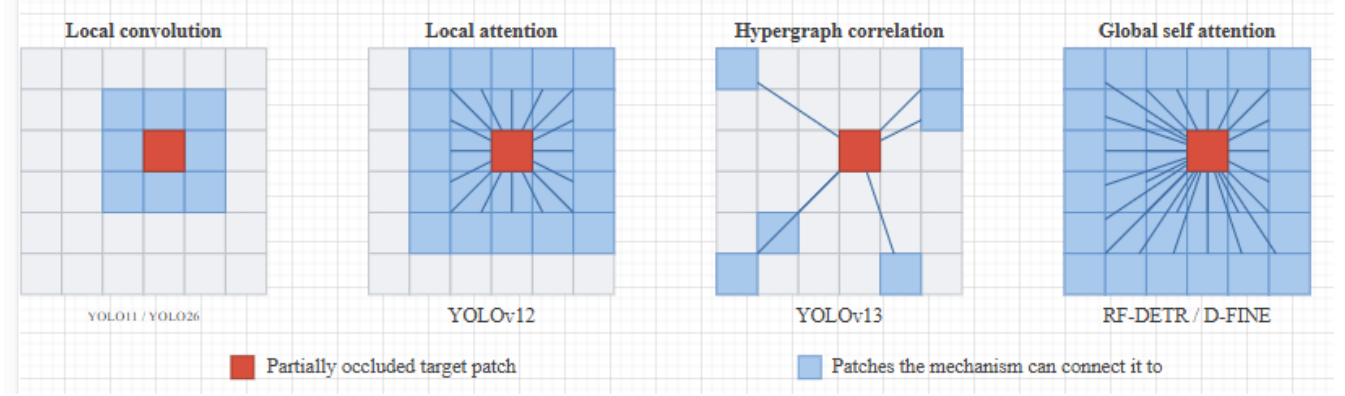


Figure 4: Detection mechanism reach across the sweep's four architecture categories of local convolution, local attention, hypergraph correlation and global self attention, constructed on Draw.io software.

State-space sequence models were considered but excluded. The Mamba lineage [19], bidirectional scanning for vision [20], YOLO - Mamba hybrid [21], and four way 2D scanning [22] offers global context with linear complexity, in retrospect a cheaper route to the same long range connections the D-FINE and RF-DETR detectors provide. Every implementation requires a custom 'selective_scan' CUDA kernel with no straightforward TensorRT or ARM64 export path, thus excluded on this deployment risk basis.

Table 2: Supporting and opposing evidence for each architecture and the inclusion decision.

Arm	Supporting evidence	Weakness / opposing evidence	Included?
YOLO11 [14]	Continuity reference and mature export path	Local convolution only, no long-range connection	Yes (Reference)
YOLO26 [14]	NMS and DFL free edge design that improves deployability without loss of accuracy	Same local only connection limitations	Yes
YOLOv12 [15]	Local attention reweights nearby evidence	Attention confined to an area of the feature map. Hence, might not link evidence across areas	Yes
YOLOv13 [16]	Hypergraph connects scattered non-adjacent evidence	Unreviewed paper	Yes
RF-DETR [17] / D-FINE [18]	Global attention, most direct fit for fragmented evidence	Higher computational cost [14] and separate repositories add export risk	Yes
Mamba lineage [19]	Linear-time scaling, cheaper than transformers	Custom selective scan kernel, so ARM64 and TensorRT export are high risk	No

2.5 Open Vocabulary Detectors (OVD)

OVDs take class names as text prompts so a held out class from training can be used for inference. Grounding DINO fuses vision and language using a transformer based detector [23] and OWLv2 scales detection training through self generated pseudo labels [24]. YOLO based models OVDs target speed with YOLO-World reports 35.4 AP at 52 fps on V100 [4] and YOLOE adds visual, text and prompt free prompting, by also surpassing YOLO-Worldv2-S by 3.5 AP. YOLOE is also $\sim 1.4x$ faster than YOLO-Worldv2-S [25].

Each model generates no text and needs one forward pass, placing OVDs between closed set detectors and VLMs for latency (Figure 5). Limitations of OVDs in this study include how reported speeds come from data centres instead of 8 GB edge machines, and accuracy is reported on general benchmarks while performance differs on modalities such as aerial imagery [26]. Therefore the performance of these models finding small, occluded targets from 5 m - 10 m is untested, making OVDs the most likely challenger to

the VLM on held out classes.

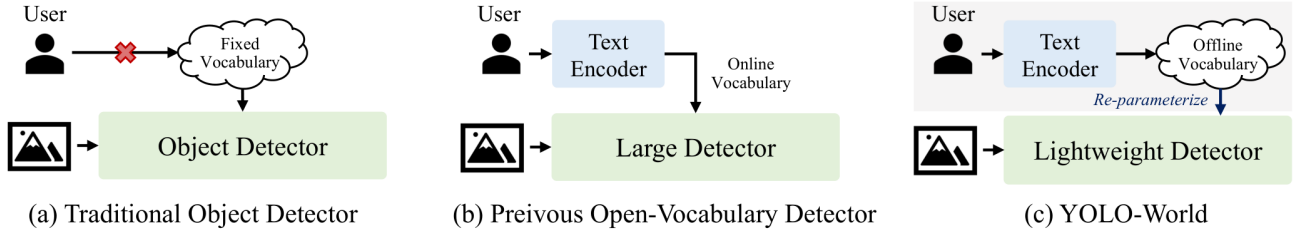


Figure 5: Three ways classes are received by a detector. (a) A traditional detector trained on a fixed vocabulary, so it can’t detect anything else. (b) Earlier OVDs encode the class names as text prompts at time of inference, needing a large detector. (c) YOLO-World encodes the prompts once into an offline vocabulary, building it into a lightweight detector, keeping inference to a single forward pass. Adapted from [4].

2.6 Vision Language Models (VLM)

VLMs can describe and answer questions about imagery but due to their size they are omitted from small platforms. MobileVLM V2’s 1.7B model matches and even beats larger 3B parameter VLMs on standard benchmarks showing the viability of ~ 2 B models [5] (Figure 6). A study by Sarwar et al. ran 5 different VLMs on a Jetson Orin Nano through Ollama and found that larger models hallucinated less and reach the first token in 18 s - 28 s against 1.5 s for a 2B model [27]. The four candidates tested here sit at the compact end of the field: SmolVLM2 is built on an architecture engineered for edge inference [28], Qwen2-VL-2B processes images at dynamic resolution [29], InternVL2.5 offers compact 1B and 2B variants [30], and Florence-2 answers detection and grounding prompts directly [31].

Onboard UAV work exists but only target guidance not detection and search reprioritise. GRaD-Nav++ runs a VLM fully onboard a drone [32], while LiteVLA-H generates action tokens in 50.65 ms and scene descriptions occur in about 150 ms - 165 ms on a Jetson AGX Orin, where decoding doesn’t produce latency which is dominated by multi-modal pre fill [33].

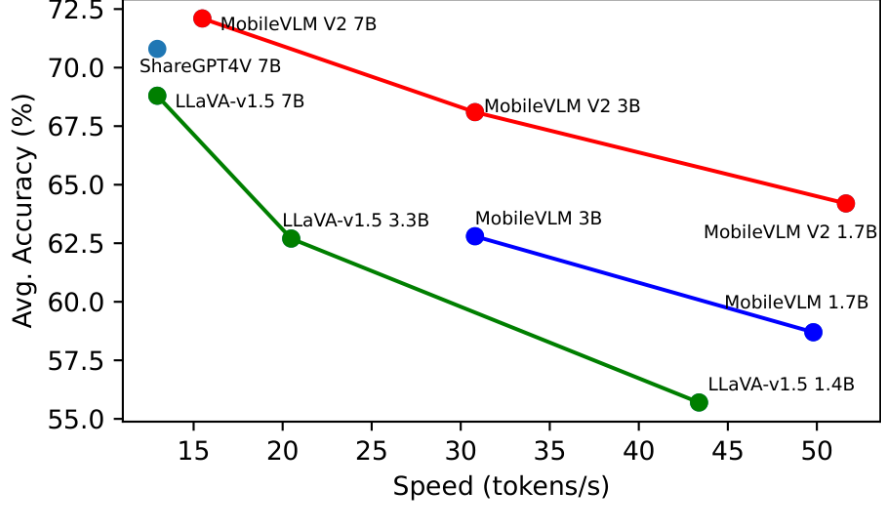


Figure 6: Average VLM benchmark accuracy against speed of generation, measured on a Jetson Orin with llama.cpp. MobileVLM V2’s 1.7B parameter model matches larger 3B models while also running faster. Adapted from [5].

None of this work verifies detector’s output and the UAV systems are navigation focused or use a larger module than Orin Nano. Pre fill dominance favours asynchronous queue interpreted on throughput rather than per frame latency. Also, a VLM relabelling what a trained detector already labelled adds latency without information, so confining the VLM role here to low confidence detections and unseen classes where it must beat cheap cropped re-detection to be considered.

Taken together, detection over open terrain is largely solved and forest occlusion is not. While onboard VLM and OVD work targets navigation or general benchmarks. None of the work reviewed combines graded occlusion, a fine HPI taxonomy, low altitude onboard detection and a fair test of VLM and OVD stages against a cropped and cheap re-detection stage under an 8 GB edge deployment constraint.

Table 3: Design decisions with supporting evidence, opposing evidence, and rebuttal.

Design decision	Supporting	Opposing	Counter Argument
RGB only sensing at 5 m - 10 m, instead of adding a new second sensor	Detection fails at ~ 300 m because too few pixels survive on occluded clues [3]. Low altitude flight restores pixels on target.	RGB thermal fusion recovers detections through sparse foliage by using contrast a RGB model can’t see [12].	Fusion assumes a second, spatially aligned and calibrated sensor. This project measures how far RGB alone goes before that cost is justified, so the fusion result sets the bar rather than invalidating the experiment.

Design decision	Supporting	Opposing	Counter Argument
Graded synthetic occlusion as the primary instrument	Masking a controlled image fraction is an established method for convolutional detectors [13]. Graded visibility shows detection collapsing well before full occlusion [2].	Synthetic masks are texturally and geometrically more crude than real canopy, and occluders placed only over targets invite shortcut learning (own analysis).	Occluders are also pasted over background during training to break the shortcut [34]. Occlusions are frozen and seeded so models get compared on identical images, and the synthetic curve is reported as a lower bound while also being cross-checked on real bush imagery.
Six architectures spanning four mechanisms, with no custom CUDA kernels created (Table 2)	Each mechanism reaches differently across an occlusion including local attention [15], hypergraph correlation [16], and global attention across the image [17, 18].	State-space models offer linear time global context at lower cost [19], while transformer pipelines need higher computational power and carry more complexity on edge hardware [14].	The exclusion is a deployment risk decision, not an accuracy judgement with every state-space implementation needing a hand-written kernel with no assured ARM64 or TensorRT path. Transformer cost is measured on-device instead of being assumed .
Cheap re-detection benchmarked before any VLM stage (baseline pipeline)	Onboard VLM latency is dominated by pre fill from prompts, not decoding [33]. On an 8 GB Jetson Orin Nano, larger models reach the first token in 18 - 28 s against 1.5 s for a 2B model [27].	Larger VLMs hallucinate less [27] and compact VLMs match 3B-scale accuracy, so the ceiling can actually be higher than latency suggests [5].	The comparison is decided by criteria that are already pre registered against re-detection rather than against nothing. A VLM that only relabels what the detector already labelled adds latency without information. Hence it is used only for low-confidence detections and unseen withheld classes.
Open-vocabulary detection as a parallel test for held-out classes from training	Text prompted detection has no token based generation and is resolved with only one forward pass[4, 25].	Zero shot accuracy degrades sharply on modalities outside of pre training, with aerial imagery among them [26]. Reported speeds also come from datacentre GPUs rather than edge devices like a Jetson Orin Nano Super.	This is why OVDs are measured here rather than assumed: on held-out HPI classes, at 5 m - 10 m, under the same Orin budget as every other configuration.

Design decision	Supporting	Opposing	Counter Argument
Onboard inference on an 8 GB module rather than offloading	Aerial deployment is constrained by communication as well as compute power, which motivates guidance (closed loop) completed on the aircraft itself [33]. Compact models also have been shown to run onboard a drone without the need for external communication or infrastructure [32].	Capable VLMs are typically cloud-based services backed by data centres [27], in addition to onboard inference being quite slow with a UAV SAR detection pipeline running at ~ 2 fps on embedded onboard hardware [9].	Cloud based inference assumes a continuous connection which is quite difficult in remote bushland and the same benchmarking work motivates local deployment on privacy and internet dependence grounds. The figure of 2 fps has also been reported without any INT8 quantisation or TensorRT acceleration application. Therefore, this thesis exports to TensorRT and reports onboard latency rather than using a raw unoptimised baseline as the upper limit.

3 Project Objectives

3.1 Research Hypothesis

It is hypothesised that running an onboard detector on a Jetson Orin Nano Super from a UAV that flies 5 - 10 m above bushland, is able to detect both persons and HPIs under varying vegetative occlusion, with mAP@50 and recall per class degrading steadily and gradually with increasing target occlusion rather than a sudden collapse, contrary to the almost total failure of detection from higher altitude over similar forest terrain [3]. By adding a secondary escalation stage, it is expected to add value only where the detector is not sufficient alone, especially on low confidence detections and withheld HPI classes from training, where a negative outcome is reported to be a valid design find for edge deployed SAR. Also, sharing detections across a multi-drone fleet with HPI and persons detections triggering different search responses throughout the fleet, reduces time to full coverage and redundant search of already visited or irrelevant locations based on detection inferences.

3.2 Research Objectives

- **O1:** By how much do precision, recall, and mAP@50 drop for person detection when a baseline HERIDAL trained detector is tested under synthetic occlusion graded from 0 to 80% occlusion level for each target on HERIDAL's own collected RGB open terrain imagery set and on freshly collected, low altitude bush dataset carrying actual vegetative occlusion?
- **O2:** Among six distinct detector candidates, which offers the best balance between accuracy, memory, local device latency and power on occluded aerial imagery when running a benchmark on the Jetson Orin? Whichever architecture wins, the available model sizes for it are swept across to characterise this tradeoff as a Pareto front, not just a single operating point.

Table 4: Detector Candidates

Model	Model Description
YOLO11s	Convolution reference and benchmark
YOLO26s	NMS free and Edge optimised
YOLOv12s	Attention within a CNN
YOLOv13s	Hypergraph correlation across non-adjacent regions
RF-DETR	Full transformer
D-FINE-S	Full transformer, refining box localisation

- **O3:** Can HPIs, including backpacks, discarded clothing, shelters plus additional classes explored during dataset construction be detected with per class recall and precision reliably enough for a human reviewer to act on alert, even when person themselves is fully occluded?
- **O4:** Does going beyond the closed-set detector (onboard VLM which verifies low confidence detections and either the VLM or an OVD handling HPI classes withheld from training) recover recall and mAP@50 without net precision loss? The Verification is benchmarked against cropped high resolution re-detections, while held out class handling compares the VLM directly to the OVD and the VLM’s processing queue stays bounded at realistic flight speed.
- **O5:** Among candidate lightweight VLMs suited to onboard deployment, which one best balances low confidence verification and classification of held out classes accurately against inference latency and memory usage the Orin Nano Super? Among OVDs, which one balances held out class detection accuracy the best against the exact same constraints
- **O6:** Does real time detection sharing across a multi-drone fleet (where HPI and person detections trigger different search responses), reduce time to full coverage and redundant area search, compared to independent, non collaborative search?

3.3 Benefits and Contribution

In terms of existing systems in the field, nothing combines low altitude onboard detection, a fine grained HPI taxonomy, graded occlusion evaluation and VLM/OVD escalation. HERIDAL reports strong results only in open terrain at 45 m - 60 m [1]. Weitefeld shows how real forest occlusion caused close to total detector failure from ~ 300 m [3]. RGB-thermal fusion views through foliage using a second sensor [12], whereas this project tests the benefits and limitations of RGB alone. For rescue operators, a system which looks deeper into the bushland extends the reachable search area which reduces the need to physically search difficult terrain.

The dataset construction, the onboard detection and classification pipeline, the VLM, OVD and detector comparisons, and the build and testing of the author’s own airframe are individual contributions. The fleet communication protocol is joint work with teammates, who each build and fly their own drones. It is included because O6 depends on it. The detection-sharing logic that runs on top of the protocol is the author’s own.

4 Methods and Methodology

4.1 Overview

This project has three concurrent lines of work converging into real world operational testing and evaluation. Firstly a detection and HPI pipeline developed on an RTX 4070 workstation then deployed on a Jetson Orin Nano Super. The second being a data workstream using published datasets now and a custom collected bushland dataset later. Thirdly, it looks into a multi drone fleet in simulation with reach towards real world fleet work depending if teammates builds are completed. The detection comparisons are put through an elimination tournament rather than a full grid (Figure 7). Six architectures, three head/backbone configurations and four pipeline configurations would need 72 separate builds for a grid based comparison, before the occlusion sweep multiplies that number again. Elimination at each stage cuts this to $\sim 30 - 40$ runs with every selection rule grounded before it's results are seen.

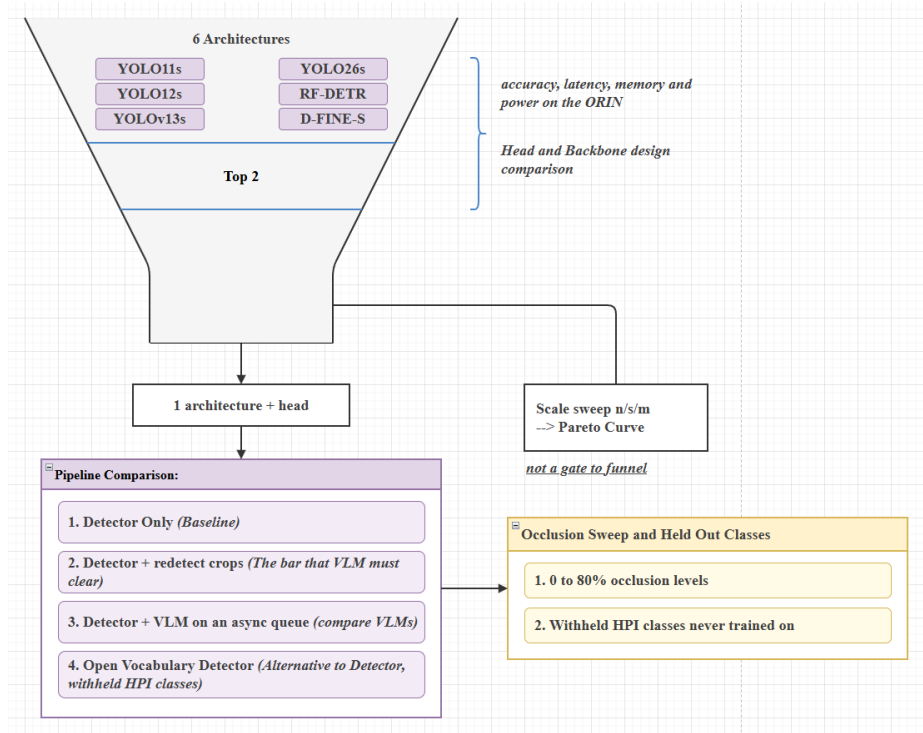


Figure 7: Elimination tournament of detector architecture comparisons. Models are ranked by mAP among those that meet the Orin's memory and latency budget. The top two architectures go through to the head comparison before the pipeline comparison, occlusion sweep and held out evaluation. The scale sweep runs separately once the architecture is known. constructed on Draw.io software.

4.2 Datasets, Split Methodology and collection constraints

Three datasets are used but never combined. HERIDAL [1] (45 m - 60 m, people only) gives the end-to-end pipeline validity and hosts the controlled occlusion sweep. It is tiled at 0.2 overlap and 1024 px, and split by source photograph. The Weitefeld dataset [3] provides real forest canopy occlusion with classes including shelter, object and person. It is split by physical finding, with three of its fifteen strips converted so far. The "unknown" class is dropped from the primary runs, using a with-and-without sensitivity check

on that class. In addition, at least 14.7% of person rows are label noise (lower bound).

HERIDAL's raw PASCAL VOC XML annotations and Weitefeld's raw "data.txt" annotations are each converted to YOLO-format labels by custom scripts, `heridal_to_yolo.py` and `weitefeld_to_yolo.py`. Both render sample tiles with boxes drawn before training, because coordinate-convention errors are invisible in overall metrics once training starts.

The bush dataset will be collected on a DJI Lito X1 drone at 5 m - 10 m while the project drone build isn't built. The training set is stills with video captures alongside for temporal work later, tagged `dji_video` and excluded from the training. It is split by placement, ideally by site with clips not spanning a split. Per HPI class, targets instances are ≥ 300 and ≥ 150 per held out class over more than 5 sessions and at least 3 different sites. The HPI taxonomy and held out class split are locked before any collection.

The DJI and Arducam AR0822 cameras have differing FOV at 82.1° and 145° respectively, thus not FOV matched at capture, revealing a scale gap with the DJI putting more pixels on a target at any given height. Some management strategies are employed post capture to mitigate this via scale augmentation, having a pixels per target comparison and being a stated limitation. The small Arducam test only batch is the sole measure of this domain gap.

The architecture sweep is trained and evaluated on bush data only, so the results are reflective of real bush conditions. HPI classes held out never enter a training set, checked and enforced by an automated script before every run (not built yet). All flight sessions are strictly conducted under Part 101 of the Civil Aviation Safety Regulations 1998 [35]

4.3 Occlusion Script

An already implemented custom occlusion script masks a set fraction of each ground truth box and logs any achieved fraction per instance. The reported mode (texture), fills the mask with vegetative occlusion from elsewhere in the same image, preserving local texture and lighting without the need for external libraries or assets unlike the flat fill of Cutout[13]. Each level of occlusion is generated once with a fixed seed (42) and frozen on disk, allowing every model to be evaluated on identical images of the same bytes. Targets occluded keep their original full box, because shrinking to visible pixels turns this into a small object study. In addition, the same occluders are pasted over background regions too during training [34], barring a detector from learning that occluders texture signals a target.

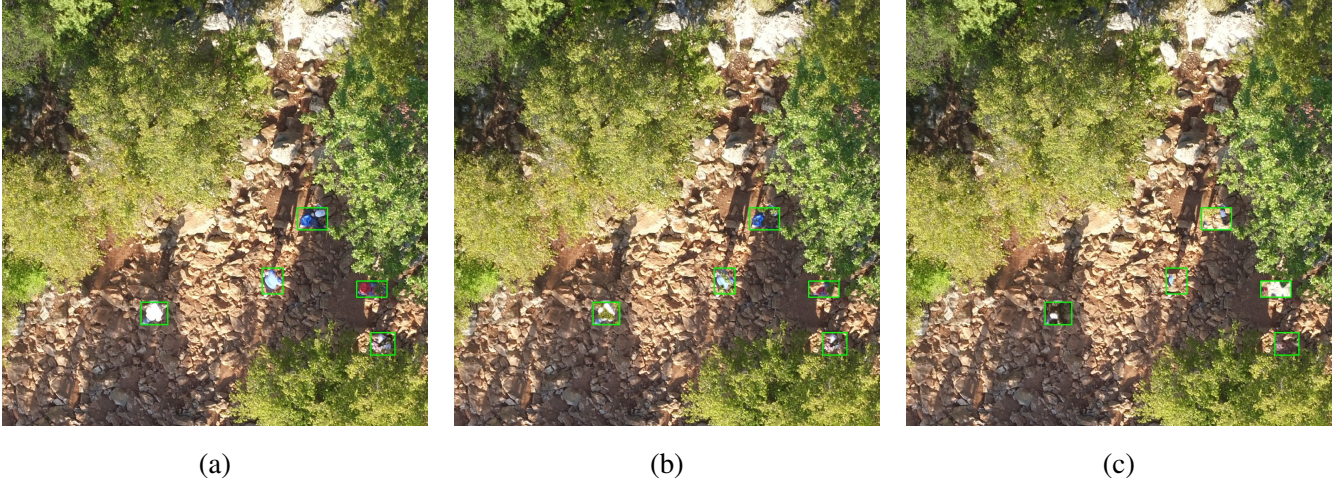


Figure 8: The exact same HERIDAL test tile at (a) 0%, (b) 40% and (c) 80% texture based occlusion after running the custom occlusion script. Full sized boxes are retained at every level.

4.4 Detector Architecture Sweep

The six architecture candidates are described in the literature review (Table 2) and Figure 9 shows the gates each one passes. One admission rule applied to all is no custom CUDA kernels for training or export since each needs to survive X86 and ARM64 compilation and TensorRT conversion. All start from pretrained weights on COCO [36] and then fine tuned on bushland data. An architecture missing the Orin’s memory and latency budget is dropped regardless of accuracy with survivors ranked by mAP on test split and top two moving forward as also shown in Figure 7.



Figure 9: Decision gates applied to each candidate architecture. Constructed on Draw.io software.

The leader then settles the training recipe of frozen backbone versus full fine-tune. LoRA, built for Transformer layers [37], doesn’t perform well when applied directly to convolutional detectors [38] and has no Ultralytics equivalent convolution variant, thus rejected. Both finalists are trained with one multi class head and a thin shared backbone feeding separate HPI and person heads, testing whether class imbalance degrades a shared head. A sweep of the winners n/s/m variants outputs the accuracy versus latency Pareto front, answering **O2**.

4.5 Pipeline Configurations and evaluation of VLMs

The detector winner is fed into four pipeline configurations with pre made adoption criteria (Table 5). Low confidence crops are re run through the same detector at higher resolution and benchmarked before any VLM usage where the VLM is measured against a cheap tuned baseline. VLMs contenders are Qwen2-VL-2B [29], SmolVLM2 [28], InternVL2.5 [30] and Florence-2 [31], all run through llama.cpp and Ollama [27].

OVDs are YOLO-World [4], YOLOE [25], Grounding DINO [23] and OWLv2 [24]. Since onboard VLM latency is pre fill dominated [33], the VLM is evaluated on how many low confidence detections it clears per second and at flight speed on queue backlog, not per frame latency.

Table 5: Pipeline Stages and Adoption Criteria

Pipeline	Adopted if
Detector only	Floor everything else must beat
Detector + re-detection ($\sim 10 - 15$ ms per crop)	Recovers meaningfully more recall within budget
Detector + asynchronous VLM queue ($\sim 2B$ parameters or fewer)	Beats the re-detection stage on held out classes and verification, with bounded backlog
Open-vocabulary detector	Displaces the VLM if it matches it at one-forward-pass cost

4.6 On Device Metrics and Evaluation

All tests run on Jetson Orin Nano Super [39] with the Arducam AR0822 (Figure 10), in MAXN SUPER mode under JetPack 6.2.2 like [27]. Clocks are locked using `jetson_clocks` since latency is clock state dependent [40]. The board is booted from an NVMe SSD with a 67 GB partition and a swap of 16 GB. TensorRT export and initial on device benchmark baselines need building still.

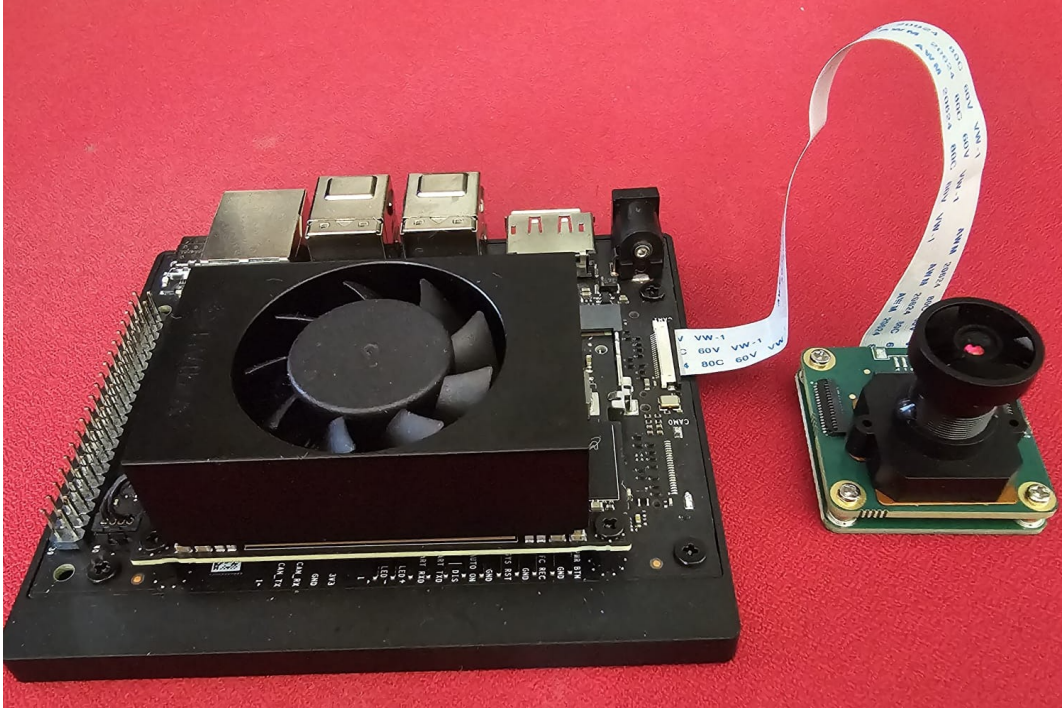
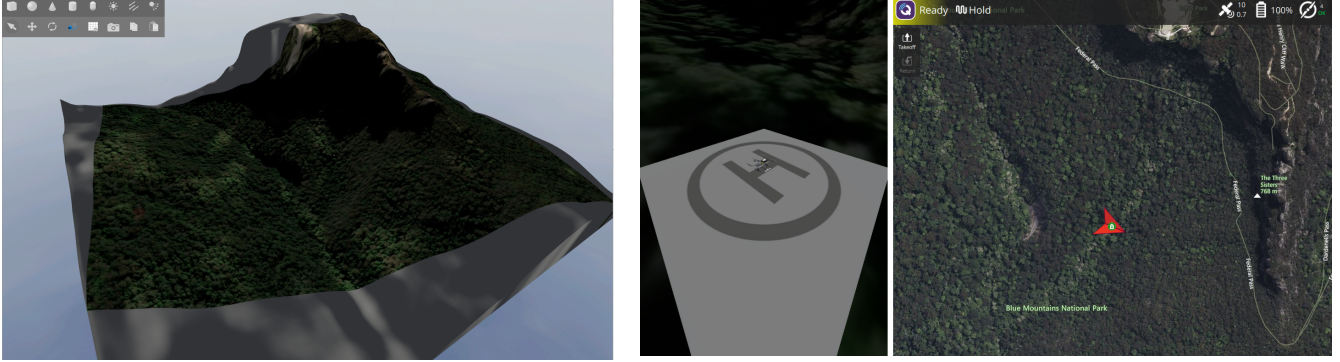


Figure 10: Onboard compute and camera: Jetson Orin Nano Super (left) and Arducam AR0822 wide-angle camera (right), connected by a ribbon cable.

4.7 Fleet Coordination

Multi-UAV search is an established SAR approach [41] developed here in simulation with PX4 SITL [42], Gazebo Harmonic [43] and a ROS2 bridge over uXRCE-DDS. The simulation is run over ROS2 Jazzy while the Jetson runs on the older ROS2 Humble so that the Arducam drivers can be used. Single drone simulation over custom real world terrain [44],[45] currently runs, as shown in Figure 11, with multi drone coordination pushed towards Semester 2 both simulation and real world. It is to be tested against mock detections in simulation, whose type field separates a person (convergence point for SAR) from a HPI (reprioritise SAR area) and evaluated on redundant re-searched area and time till full coverage.



(a) Gazebo Harmonic terrain of Blue Mountains National Park, built from satellite imagery and elevation data.

(b) PX4 x500 Quadcopter model on its launch pad in Gazebo (left), and the vehicle state in QGroundControl (right).

Figure 11: Single-drone simulation with custom real world terrain.

4.8 Resources

Model training is conducted on a local RTX 4070 machine with the Ultralytics framework [46] for YOLO11, YOLO26 and YOLOv12 and uses author's own repositories for YOLOv13, RF-DETR and D-FINE. The Jetson Orin Nano Super was chosen for its CUDA/TensorRT support and prior use cases in UAV SAR [9]. The Jetson also peaks at 67 sparse INT8 TOPS and 33 dense [47] while a Raspberry Pi 5 reaches 13 or 26 TOPS with the AI HAT+ add on attached [48] with both values being manufacturer claimed figures and not throughput. Bush dataset collection uses a DJI Lito X1 drone and Arducam AR0822 on custom airframe with annotation using CVAT, and the Arducam AR0822 on the custom airframe provides the small test batch only.

5 Timeline and Risk Management

The timeline is presented in Appendix A as a Gantt chart. The critical path runs from custom dataset collection through detector evaluation, pipeline integration and field trials to results analysis, so a collection delay postpones everything after it. Key milestones include dataset collection (16 Nov), detector evaluation (23 Nov) and VLM evaluation (30 Nov). Data collection was the only task to fall behind so it was extended from six to nine weeks. The baseline datasets don't need collected data so current work isn't blocked. The risk Table 6 lists the primary risks, their likelihood, consequences and management strategies in place. No significant environmental risk is in line with this project and the only financial risk is a rise in prices of parts.

Table 6: Risk Analysis.

Risk	Likelihood	Consequence	Mitigation
UAV flight at 5 to 10 m near people	Low	High	Part 101 operating rules [35], landholder permission, object-only sessions first, standard pre-flight checks
Weather or logistics delay bush collection (critical path)	Medium	High	Nine week collection window, staged sessions annotated as each finishes, Weitefeld fallback, DJI used so the airframe is not required
Airframe not flight-ready	Medium	Medium	Simulation and bench footage fallback
Separate-repo architectures fail TensorRT or ARM64 export	Medium	Medium	Admission rule, export tested early, three Ultralytics native architectures ensure a comparison is still possible
The 30 to 40 planned training runs exceed the single RTX 4070's available time	Medium	Medium	The sweep is an elimination tournament, not a full grid, where only the top two architectures carry forward. Each yolo11s baseline took between 2.8 and 5.1 hours, so 30 to 40 runs come to approximately 85 to 205 GPU hours. Those times are from yolo11s and the transformer architectures will probably run slower. There is no shared queue to wait in for usage of the GPU (RTX 4070), and the sweep starts on staged bush data instead of waiting for collection to finish.
8 GB memory makes the VLM infeasible	Medium	Low	~2B parameter cap, quantisation, NVMe swap, asynchronous queue, a negative result is valid
Held-out class leaks into training	Low	High	An automated check, still needing to be built and used before every training run, will abort if any withheld class appears in any training
DJI to Arducam domain gap	High	Medium	Arducam test-only batch, scale augmentation, pixels-per-target comparison, stated limitation
Weitefeld test split too small to support claims (2 findings)	High	Medium	Convert the remaining strips, bootstrap confidence intervals over findings, limitation reported
Loss of data	Low	High	Code and results in git, datasets checked through provenance files, bush data copied to a second location after each session
Teammates' drones or fleet communication not ready for O6	Medium	Low	Fleet coordination is tested in simulation against mock detections with real-world fleet trials treated as an extension and O1 to O5 don't depend on teammates' work

6 Progress to Date

Progress is on track with the revised Gantt Chart. Training environment, baseline tests, dataset converters, occlusion script, Jetson setup and testing, and a single drone simulation stack are all completed. Test set evaluation harness, full scale occlusion sweeps and TensorRT baseline remain.

The first Weitefeld overfit gate scored 0.31 mAP@50 since the paper omits the y axis direction, shifting every box up by its height. After this correction, the 3 class gate passed at 0.861 and since 14.7% of person rows are marked "excluded, not checked", person metrics are a lower bound.

On validation split (Table 7), HERIDAL reached 0.954 mAP@50, against 0.802 mAP@50 [9] and 88.9% recall at 34.8% precision [1], although evaluation protocols differ. Weitefeld person reached 0.331 mAP@50 where its authors' YOLOv12 failed [3]. Its recall never exceeded 0.55 (Figure 13) and shelter and object fail despite ample training boxes. The converted scripts hold only 47 findings (Table 8) so more converting needed before any test claim

Table 7: Baseline results on the validation split.

Baseline	Dataset	Class	Val boxes	P	R	mAP@50	mAP@50-95
A	HERIDAL	person	458	0.922	0.900	0.954	0.641
B	Weitefeld	person	248	0.705	0.226	0.331	0.114
B	Weitefeld	object	65	0.090	0.123	0.019	0.005
B	Weitefeld	shelter	258	0.000	0.000	<0.001	<0.001

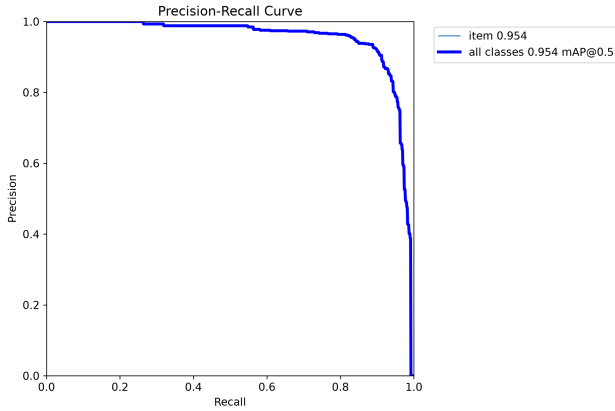


Figure 12: Precision versus recall curve on the validation split of Baseline A (HERIDAL)

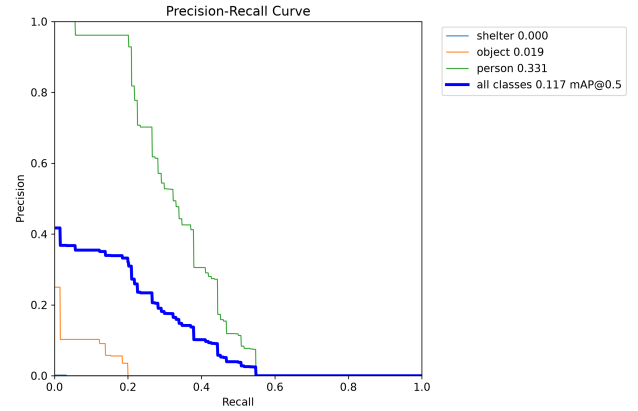


Figure 13: Precision versus recall curve on the validation split of Baseline B (Weitefeld)

Table 8: Weitefeld 3 class split composition (three out of fifteen strips).

Split	Tiles	Shelter	Object	Person	Findings
Train	3,850	805	424	730	36
Validation	1,002	258	65	248	9
Test	308	0	54	57	2

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A Project Gantt Chart

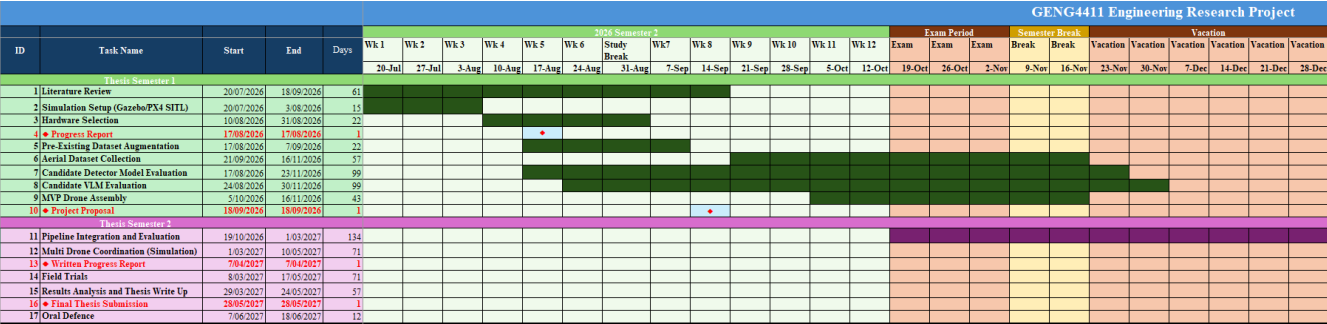


Figure 14: Project Gantt chart for Semester 1 of Thesis

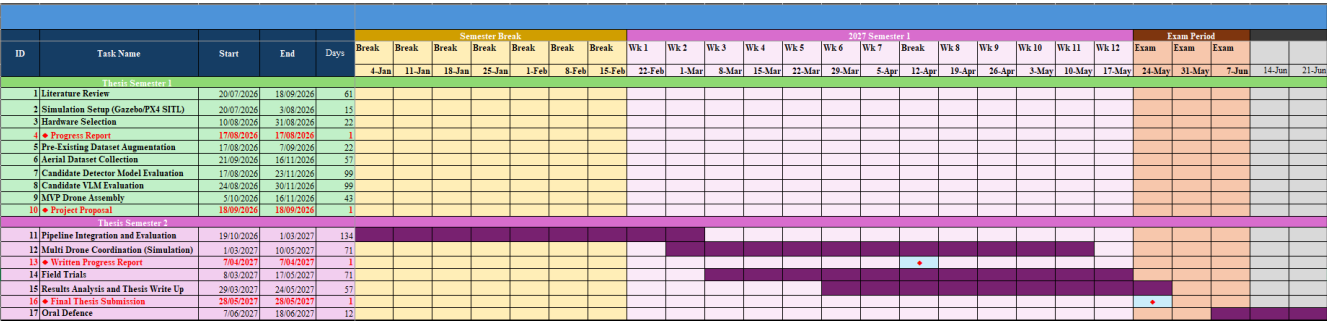


Figure 15: Project Gantt chart for Semester 2 of Thesis